

# **EC200U&EG91xU Series**

# **QuecOpen(SDK) ADC**

# **Development Guide**

**LTE Standard Module Series**

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# About the Document

## Revision History

| Version | Date       | Author   | Description                                                                                                                                                                                                     |
|---------|------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -       | 2021-10-08 | Neo KONG | Creation of the document                                                                                                                                                                                        |
| 1.0     | 2021-10-19 | Neo KONG | First official release                                                                                                                                                                                          |
| 1.1     | 2023-10-20 | Sum LI   | <ol style="list-style-type: none"><li>1. Added applicable modules EG912U-GL and EG915U series.</li><li>2. Updated the description related to log capture before ADC function debugging (Chapter 3.3).</li></ol> |

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# 1 Introduction

Quectel EC200U and EG91xU family (EG912U-GL and EG915U series) modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS, which is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]**.

This document is applicable to QuecOpen® solution based on CSDK build environment. It introduces the ADC function of Quectel EC200U and EG91xU family modules, including the ADC API and development process in QuecOpen® solution.

## 2 ADC API

### 2.1. Header File

*ql\_adc.h*, the header file of ADC API, is in the `components\ql-kernel\inc` directory. Unless otherwise specified, all the header files mentioned in this document are in this directory.

### 2.2. API Description

#### 2.2.1. ql\_adc\_get\_volt

This function reads the analog voltage value from the ADC channel.

- **Prototype**

```
ql_errcode_adc_e ql_adc_get_volt(int ql_adc_channel_id, int *adc_value)
```

- **Parameter**

*ql\_adc\_channel\_id*:

[In] The specified ADC channel. See **Table 1** for the correspondence between this ADC channel and the actual ADC physical channel.

*adc\_value*:

[Out] Measured voltage value. Unit: mV.

- **Return Value**

See **Chapter 2.2.1.1** for error codes.



**Table 1: The Correspondence Between ADC Channel and the Actual ADC Physical Channel**

| ADC Channel     | Actual ADC Physical Channel |
|-----------------|-----------------------------|
| QL_ADC0_CHANNEL | ADC0                        |
| QL_ADC1_CHANNEL | ADC1                        |
| QL_ADC2_CHANNEL | ADC2                        |

**NOTE**

EG912U-GL and EG915U series modules only support ADC0 and ADC1.

### 2.2.1.1. ql\_errcode\_adc\_e

The enumeration of error codes is defined as follows:

```
typedef enum
{
    QL_ADC_SUCCESS = 0,
    QL_ADC_INVALID_PARAM_ERR = 10 | (QL_COMPONENT_BSP_ADC << 16),
    QL_ADC_GET_VALUE_ERROR = 50 | (QL_COMPONENT_BSP_ADC << 16),
    QL_ADC_MEM_ADDR_NULL_ERROR = 60 | (QL_COMPONENT_BSP_ADC << 16),
} ql_errcode_adc_e;
```

- **Member**

| Member                     | Description           |
|----------------------------|-----------------------|
| QL_ADC_SUCCESS             | Successful execution. |
| QL_ADC_INVALID_PARAM_ERR   | Invalid parameter.    |
| QL_ADC_GET_VALUE_ERROR     | Failed execution.     |
| QL_ADC_MEM_ADDR_NULL_ERROR | Null pointer address. |

# 3 ADC Development Process

This chapter introduces how to use ADC-related API functions in applications and perform simple tests.

## 3.1. ADC Initialization

To test the ADC example, you need to uncomment `ql_adc_app_init()` (as shown in the following figure) to initialize ADC. This function is disabled by default and is located in `ql_init_demo_thread()` in `\components\ql_application\init\ql_init.c` directory.

```
static void ql_init_demo_thread(void *param)
{
    QL_INIT_LOG("init demo thread enter, param 0x%x", param);

    ql_audio_app_init();
    ql_lcd_app_init();
    ql_nw_app_init();
    //ql_gpio_app_init();
    //ql_gpioint_app_init();
    //ql_datacall_app_init();
    ql_osi_demo_init();
    ql_dev_app_init();
    ql_adc_app_init();

    //ql_http_app_init();

    ql_rtos_task_sleep_ms(10);
    ql_rtos_task_delete(NULL);
}
```

## 3.2. Testing ADC in Applications

If you need to test the ADC in applications, you can refer to the example in the following path:  
`\components\ql_application\peripheral\adc_demo.c`.

The functions in `adc_demo.c` is parsed as follows:

- `ql_adc_demo_test()`: Execution function of a task for reading the voltage value from the ADC channel, and you can modify the code based on your needs.
- `ql_adc_demo_thread()`: ADC demo thread for calling `ql_adc_demo_test()`.
- `ql_adc_app_init()`: create and add a task to the initialization queue.

### 3.3. ADC Function Debugging

ADC function of EC200U and EG91xU family modules can be debugged on Quectel LTE OPEN EVB.

You can read the voltage value through USB AP Log Port after capturing the log with *cooltools*. For details about log capture, see **document [2]**.

# 4 Appendix References

**Table 2: Related Documents**

| Document Name                                                    |
|------------------------------------------------------------------|
| [1] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Quick_Start_Guide |
| [2] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Log_Capture_Guide |

**Table 3: Terms and Abbreviations**

| Abbreviation | Description                   |
|--------------|-------------------------------|
| ADC          | Analog-to-digital Converter   |
| API          | Application Program Interface |
| IoT          | Internet of Things            |
| RTOS         | Real-Time Operating System    |
| SDK          | Software Development Kit      |
| USB          | Universal Serial Bus          |